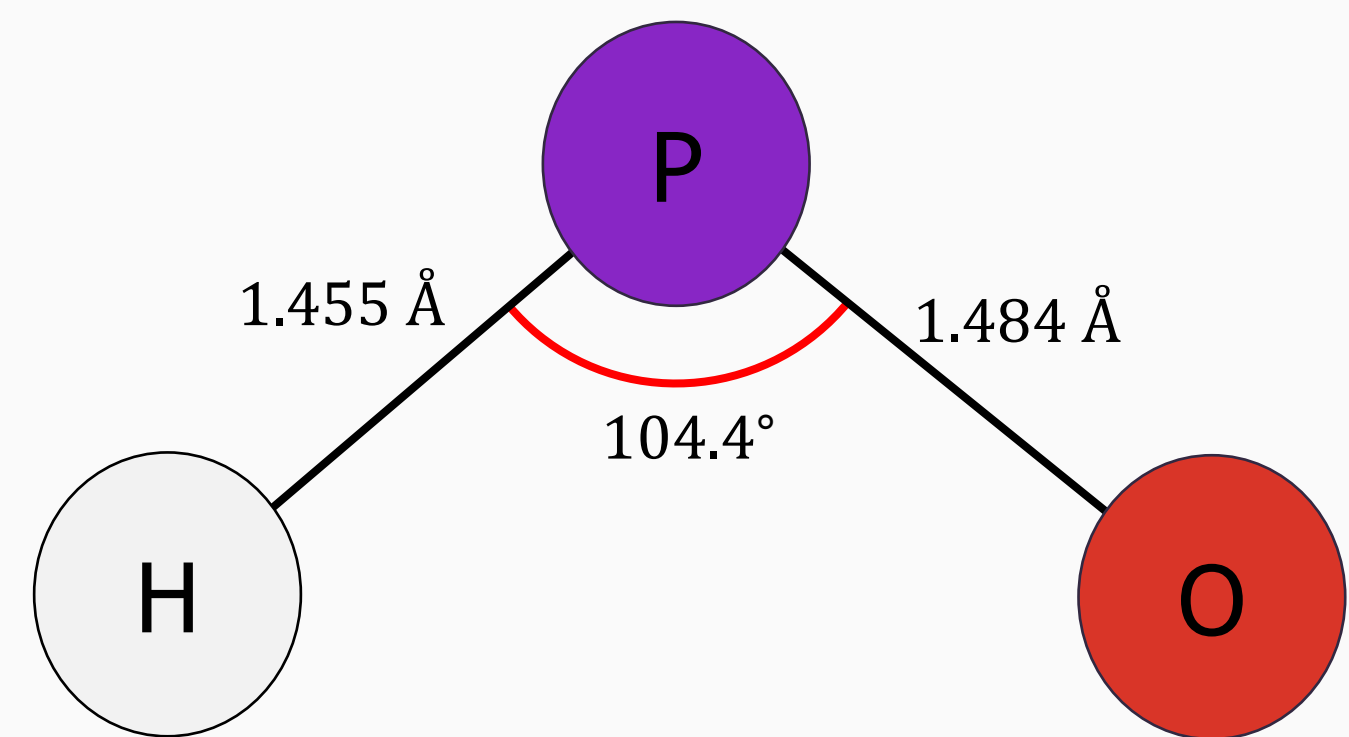


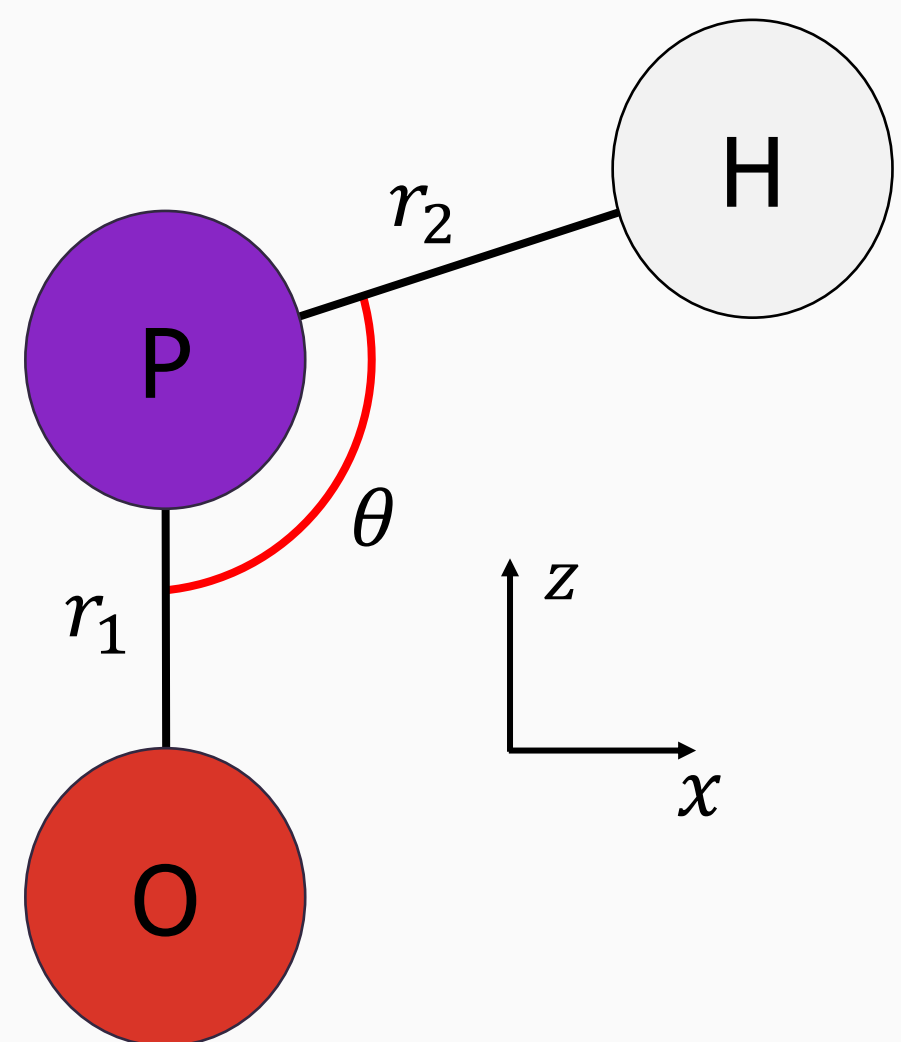
Motivation: Why HPO?

HPO appears as an intermediate molecule in proposed abiotic mechanisms to form phosphine on Venus.

Equilibrium stretch and angles at the electronic ground state:

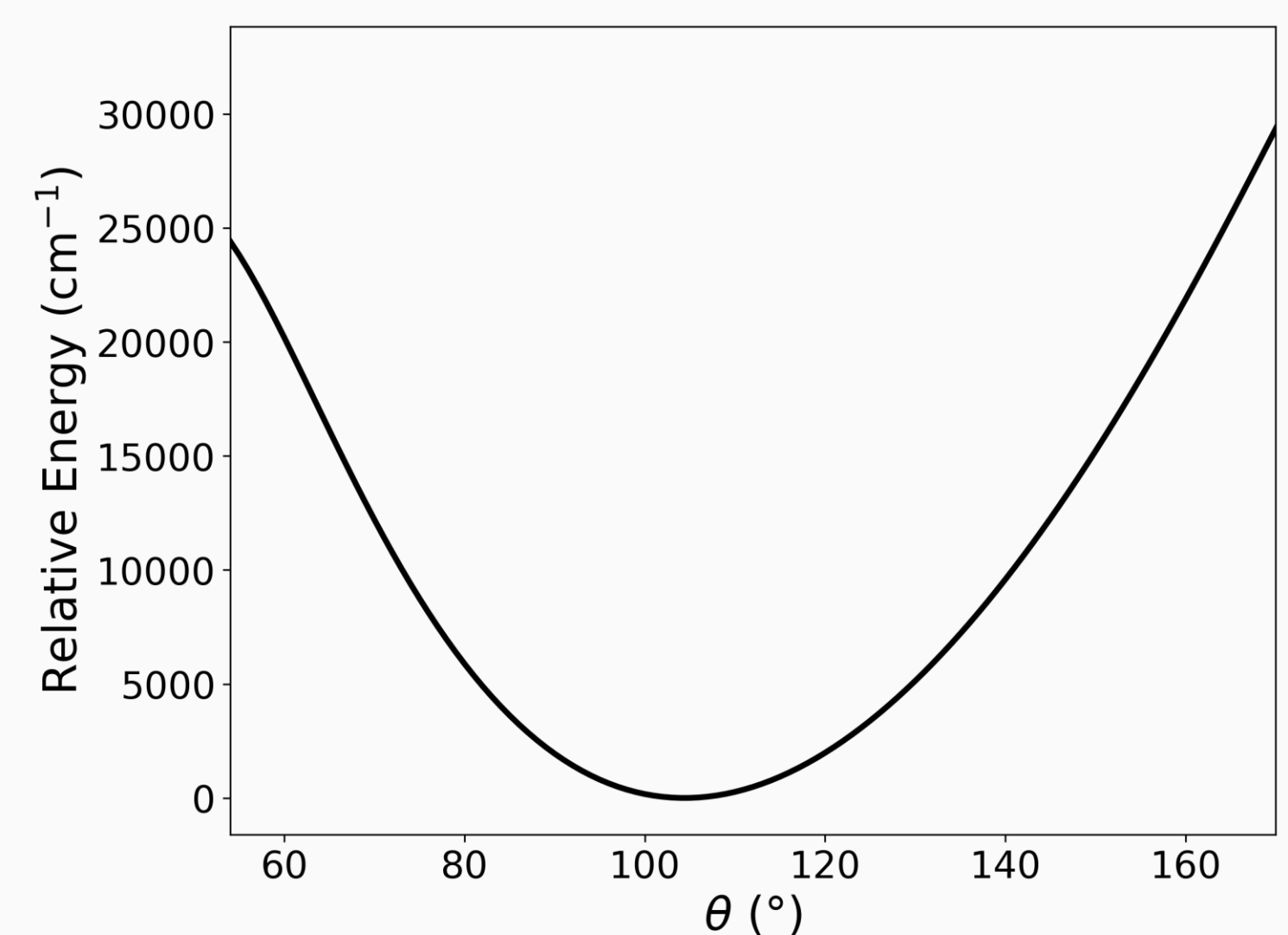
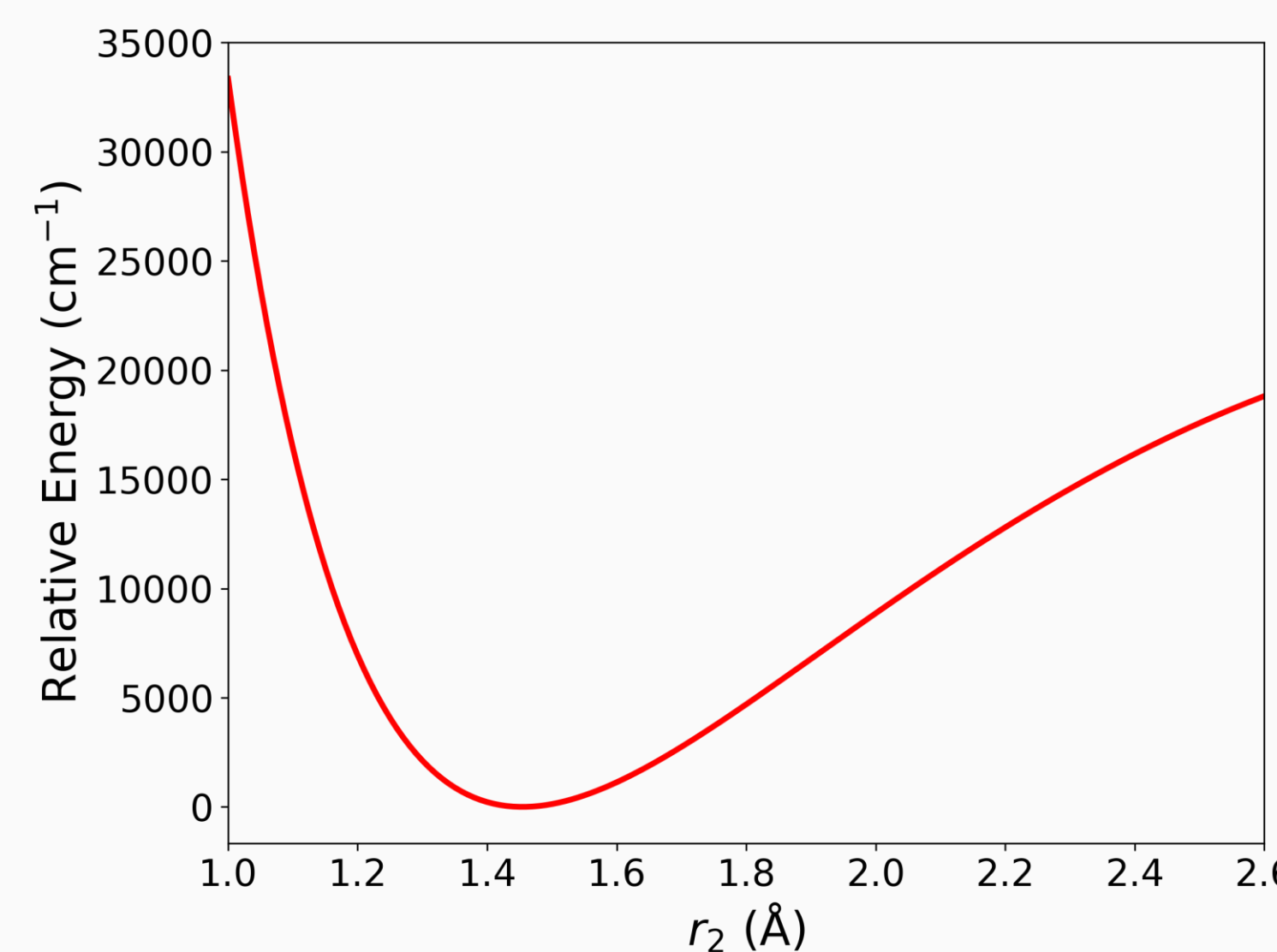
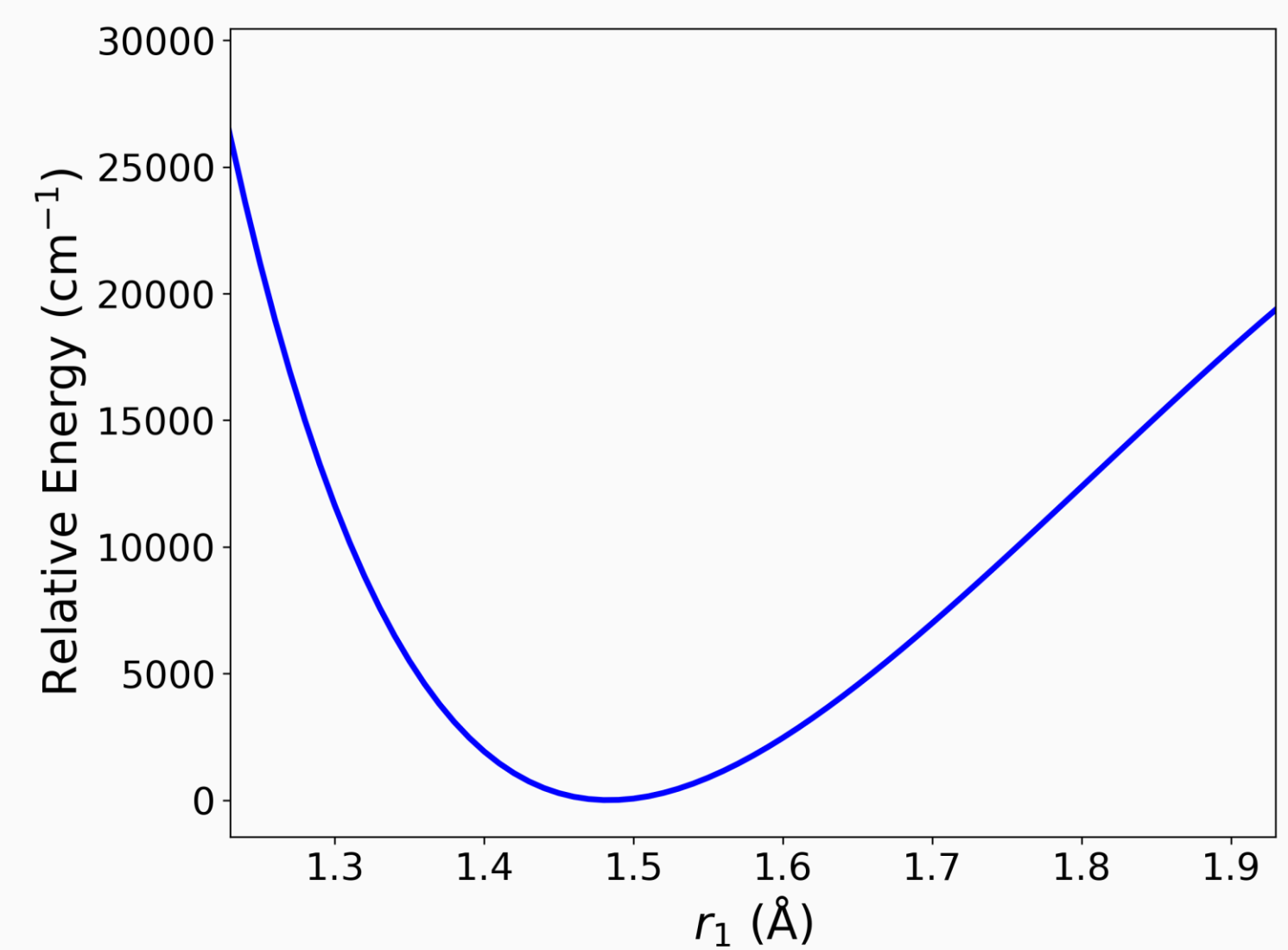


We use the internal coordinates and cartesian axis as presented:



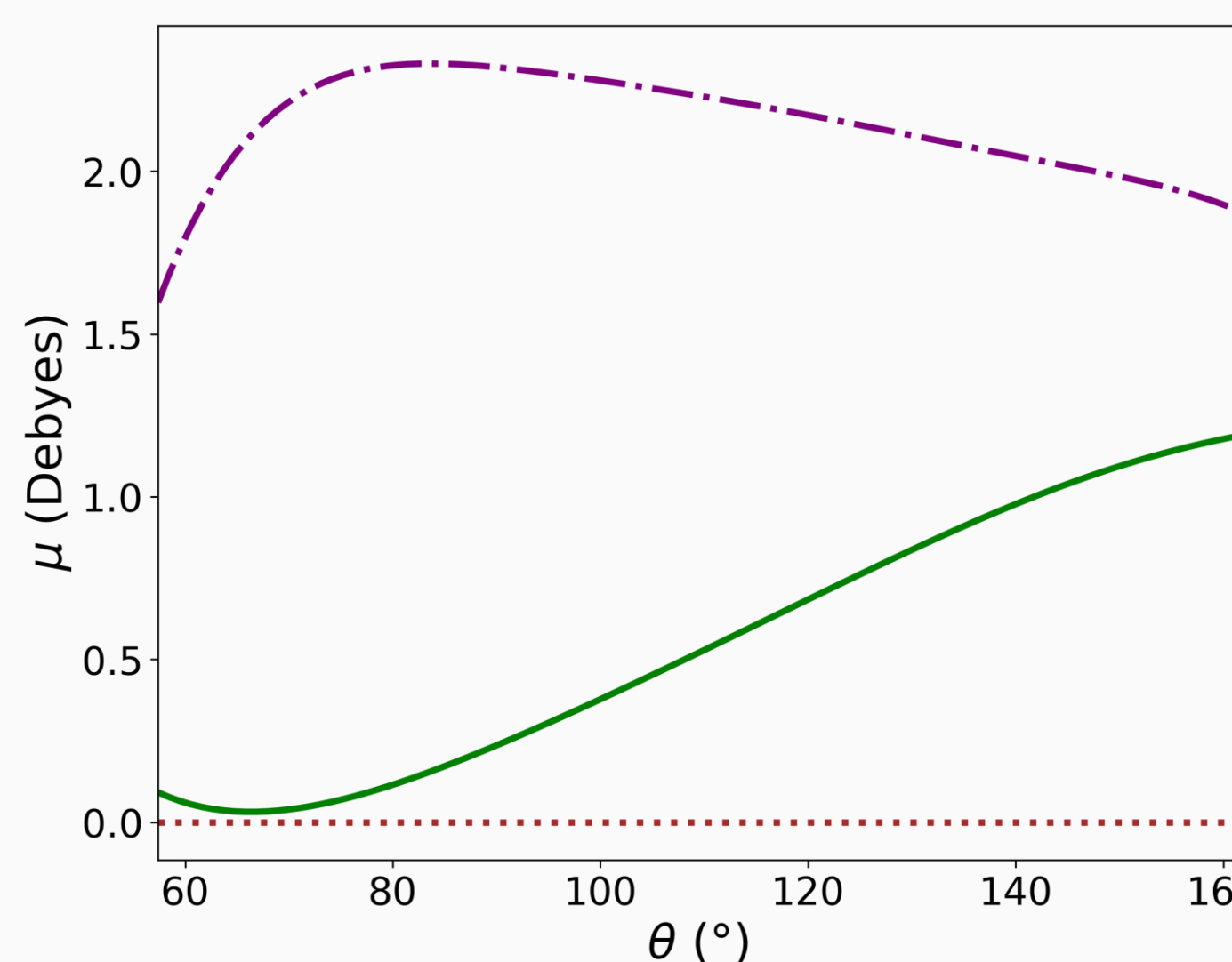
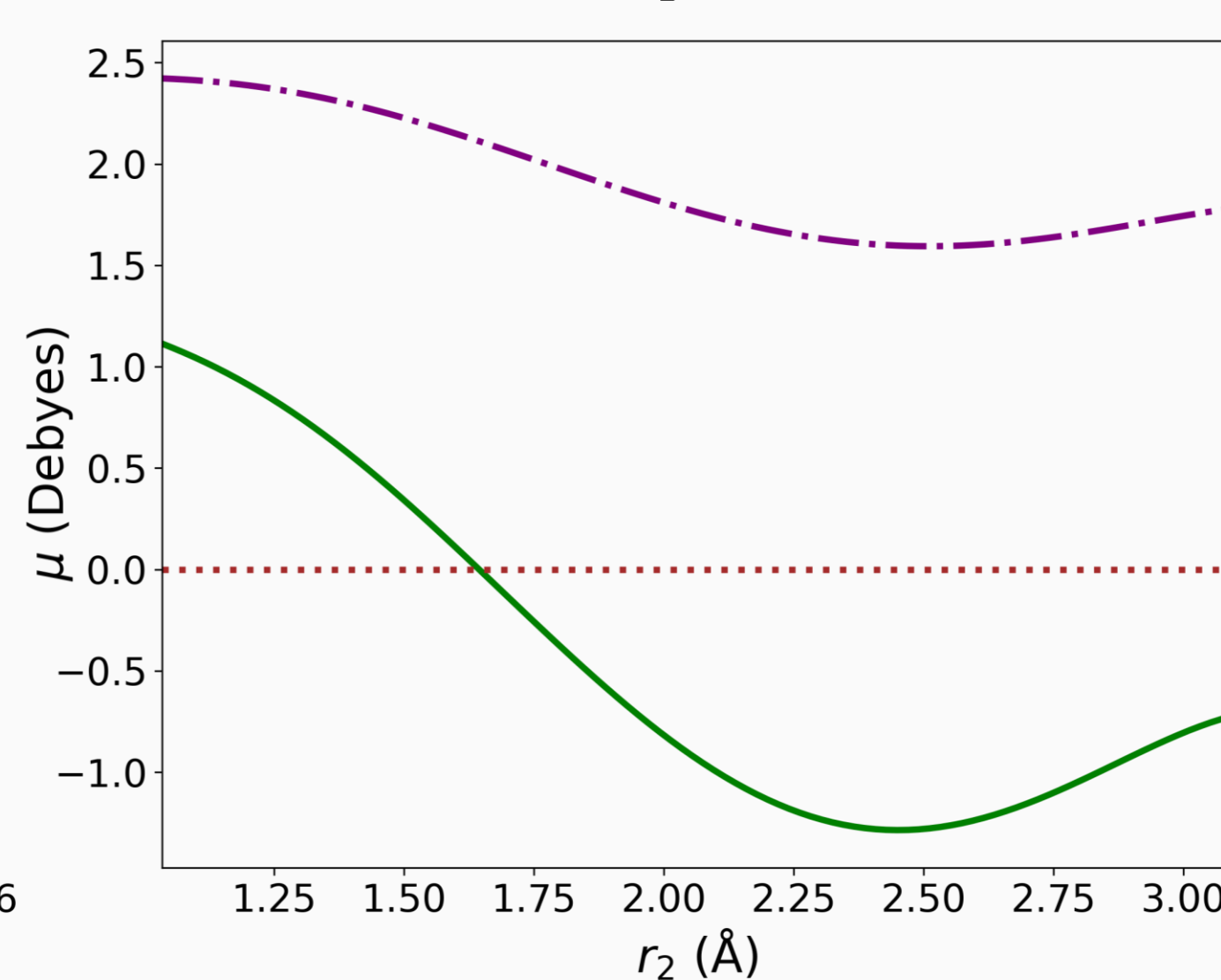
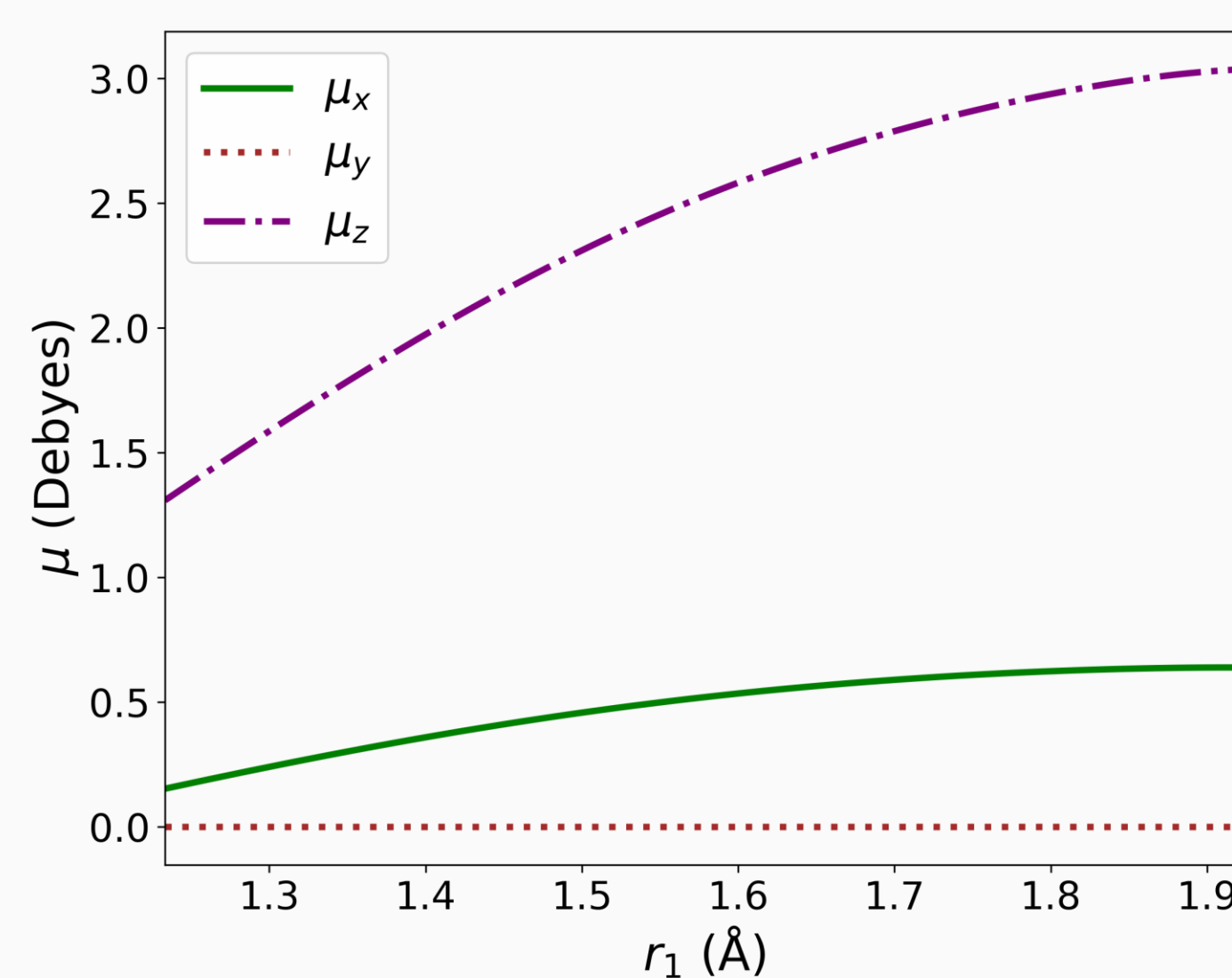
Potential Energy Surface

7770 *ab initio* points computed with CCSD(T)-F12/cc-pVQZ-F12a



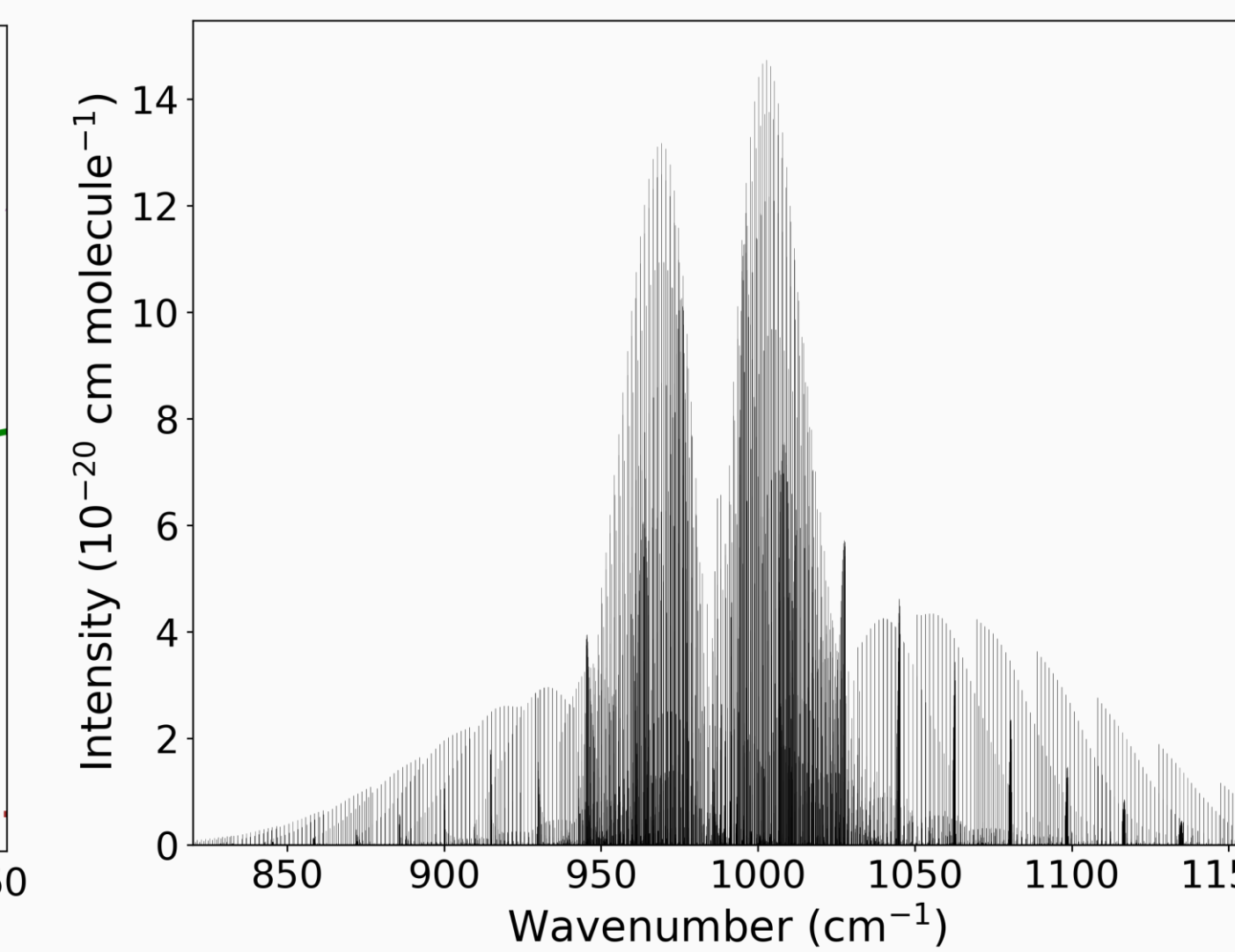
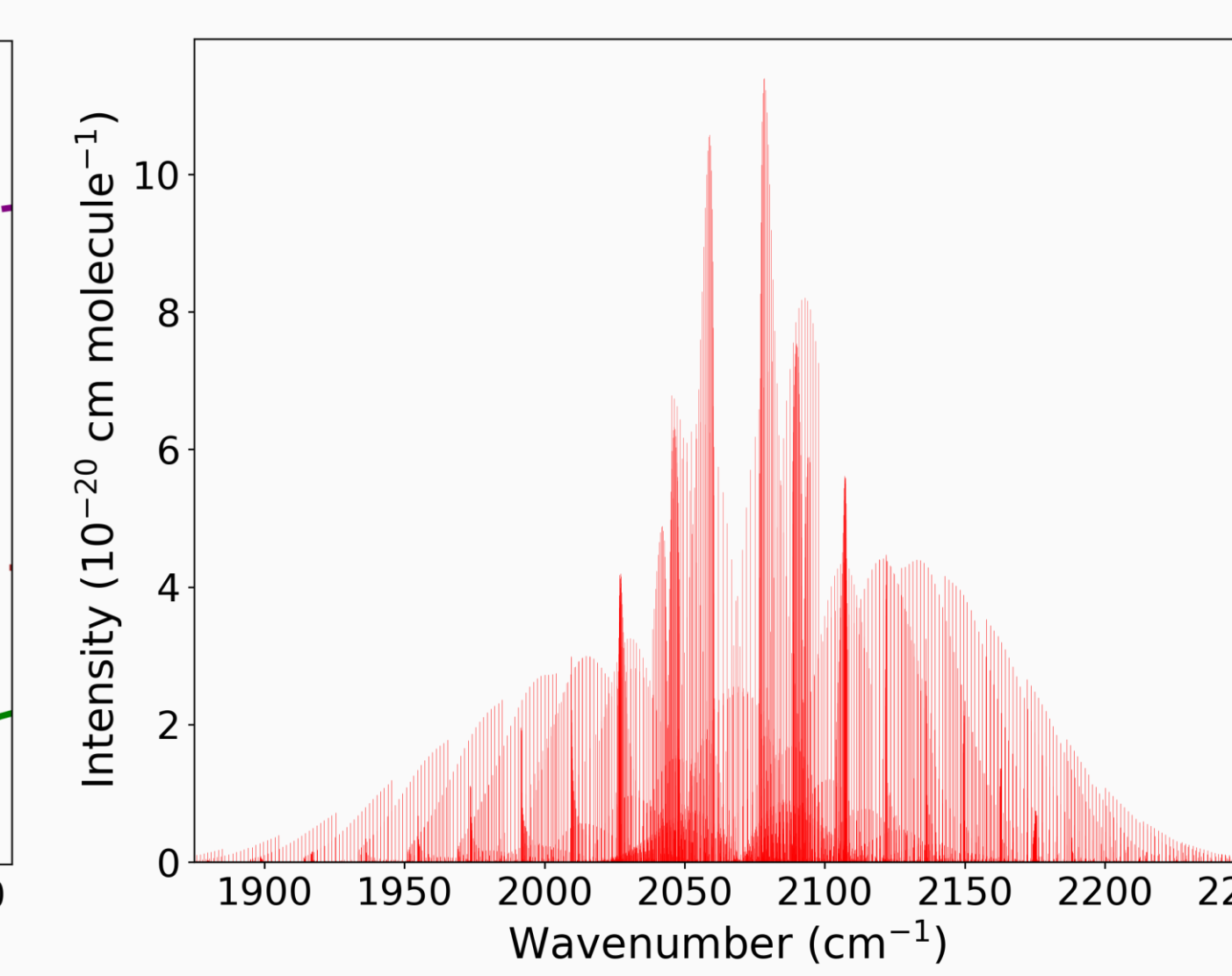
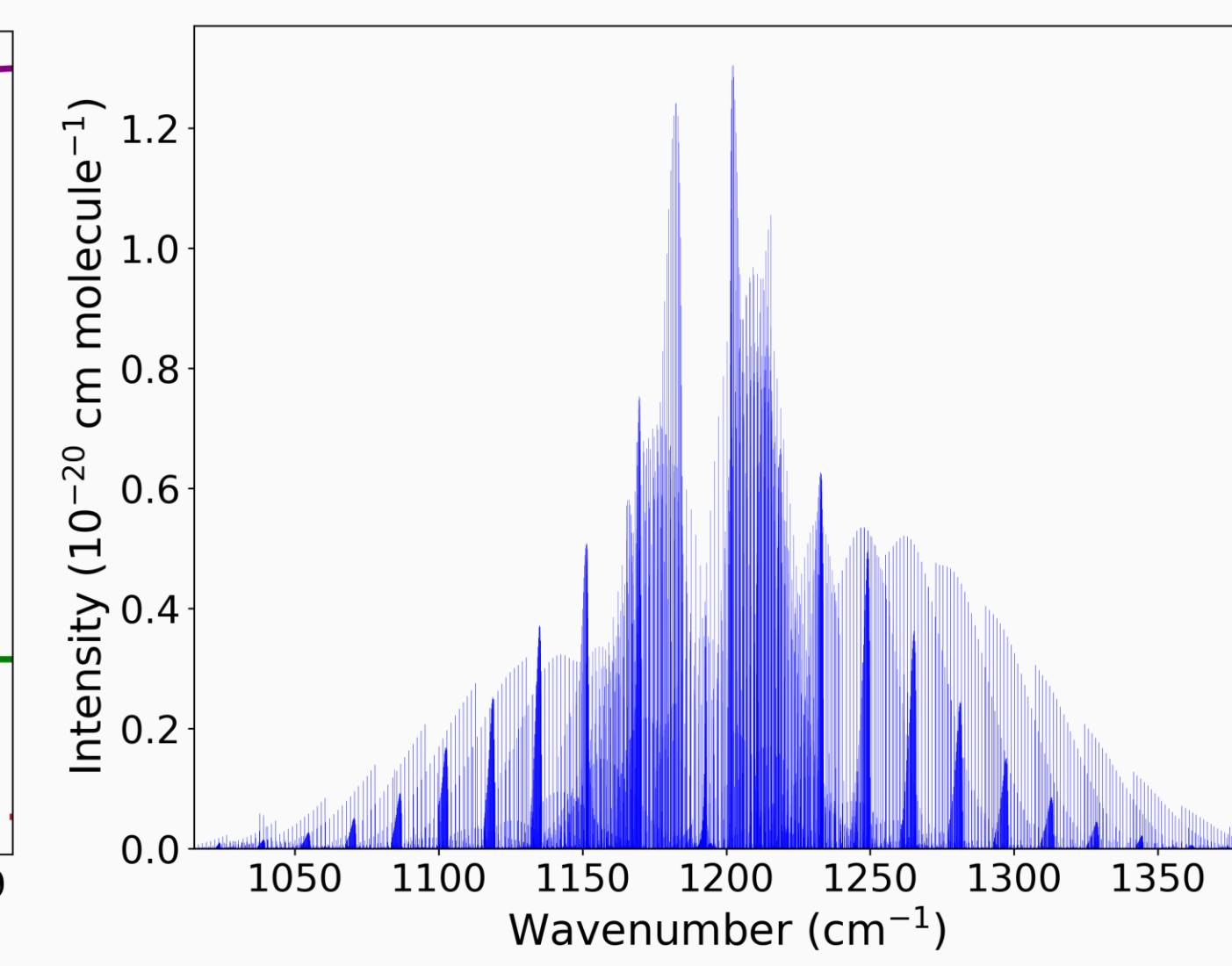
Dipole Moment Surfaces

Same grid as the PES computed with CCSD(T)/aug-cc-pV(5+d)Z

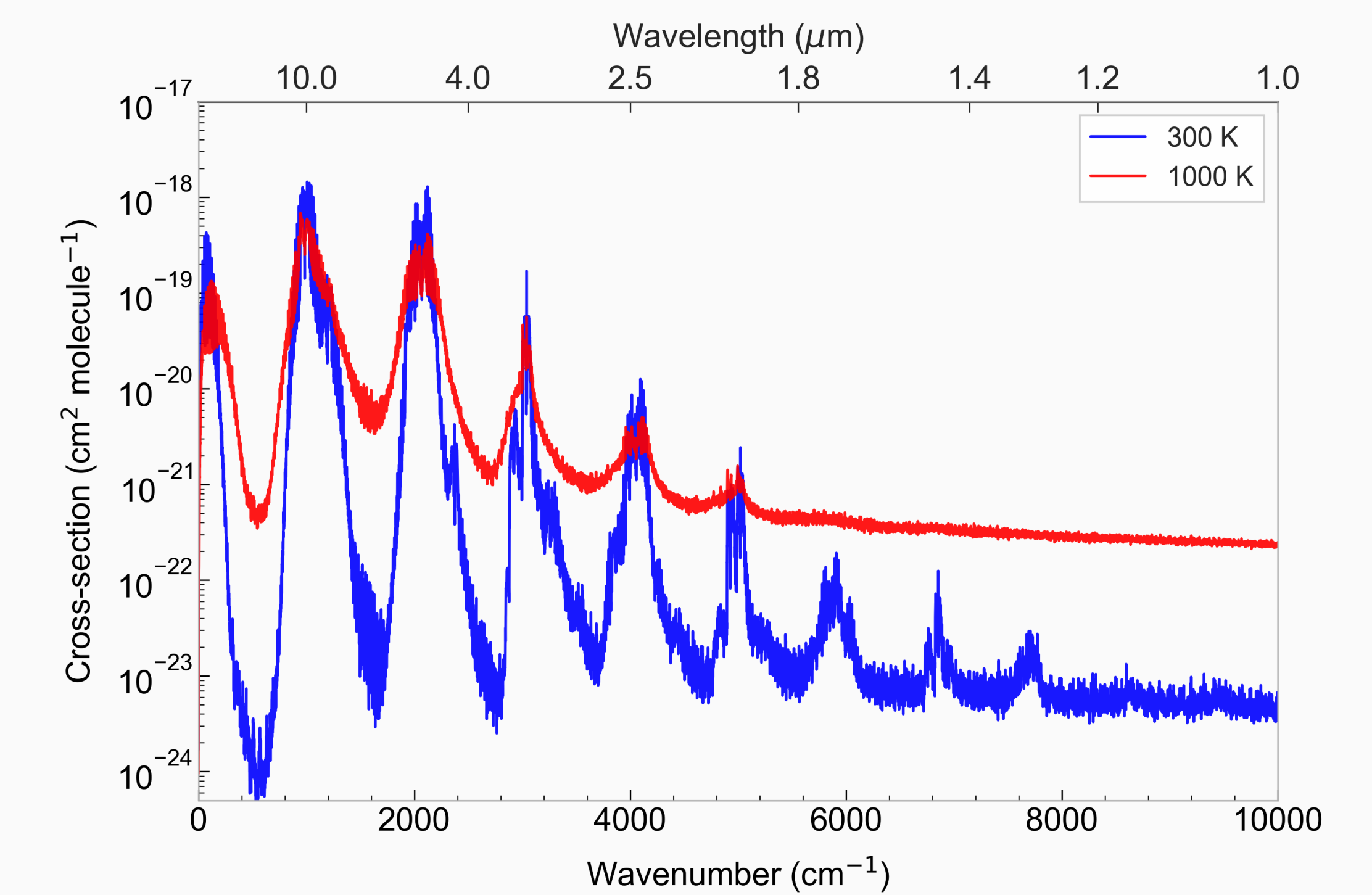


Stick Spectra

Rotational stick spectra of the fundamental vibrational modes



Cross-sections



- Fitted PES has root-mean-square error of 2 cm⁻¹ for energies under 18,000 cm⁻¹
- Molecule appears relatively rigid (stiff bending), we can use non-singular methods to perform variational calculations.
- Dipole values calculated oriented with respect to the cartesian axis given.
- Fitted to a Taylor-series type expansion.
- The first and third fundamental modes have been experimentally measured, but not the second, despite intensity seeming higher.